



Forecasting Day-Ahead LMP

Ping Zhong • Energy Analyst II 3rd Stage Interview • June 2026

Agenda

- 1 The problem & the data
- 2 Feature engineering
- 3 Model selection
- 4 Performance
- 5 Takeaways & what's next

Why forecast day-ahead prices?

A day-ahead forecast is a position decision. The value is not just the number — it's knowing when to trust it.

01

Power Traders

Day-ahead vs. real-time positioning. The DA/RT spread is where P&L is made or lost.

02

Asset Owners

Dispatch for generation and storage against expected prices.

03

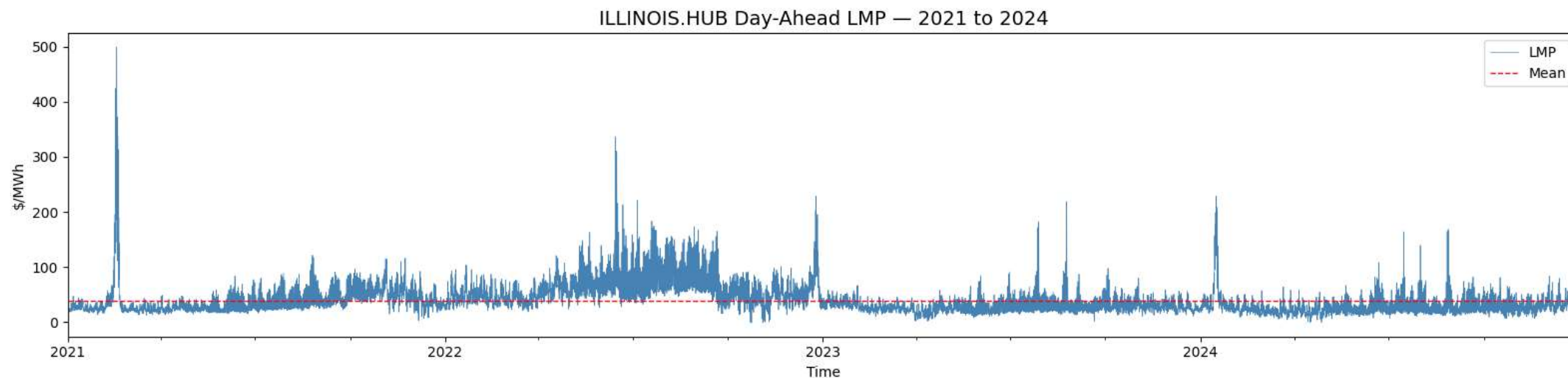
Risk Desks

Sizing exposure and stress-testing portfolios against price spikes.

MISO Illinois Hub — a liquid, well-traded hub in the largest US wholesale market by footprint.

What we're predicting

A calm baseline punctuated by violent spikes. The spikes are rare — and they are where the money and the risk live.



~35,000 hrs

2021–2024, hourly

Train 22–23

Test on 2024

33 features

Temporal · lags · rolling

5 hrs < \$0

Negatives kept, not dropped

33 features — but a few do most of the work

Temporal patterns, autoregressive lags, rolling statistics, and market signals. The lags carry the signal.

Group	Features	Count
Target	LMP	1
Components	Energy, Congestion, Loss	3
Temporal (raw)	hour, day_of_week, month, quarter	4
Temporal (flags)	is_weekend, is_monday, is_summer, is_winter, is_peak_hour, is_offpeak_hour	6
Temporal (circular)	hour_sin, hour_cos, dow_sin, dow_cos, month_sin, month_cos	6
Lagged LMP	lmp_lag_24h, lmp_lag_48h, lmp_lag_72h, lmp_lag_168h	4
Lagged Congestion	congestion_lag_24h, congestion_lag_168h	2
Spike flag	spike_lag_24h	1
Rolling stats	lmp_roll_24h_mean, lmp_roll_48h_mean, lmp_roll_7d_mean, lmp_roll_24h_std, lmp_roll_7d_std, lmp_roll_24h_max, lmp_roll_24h_min, lmp_vs_7d_avg	8
Holidays	is_holiday, is_day_before_holiday	2

Why XGBoost

On average error, every model ties within ~3%. The choice is about interpretability and tail behavior — not raw accuracy.

- **Accuracy is a wash**

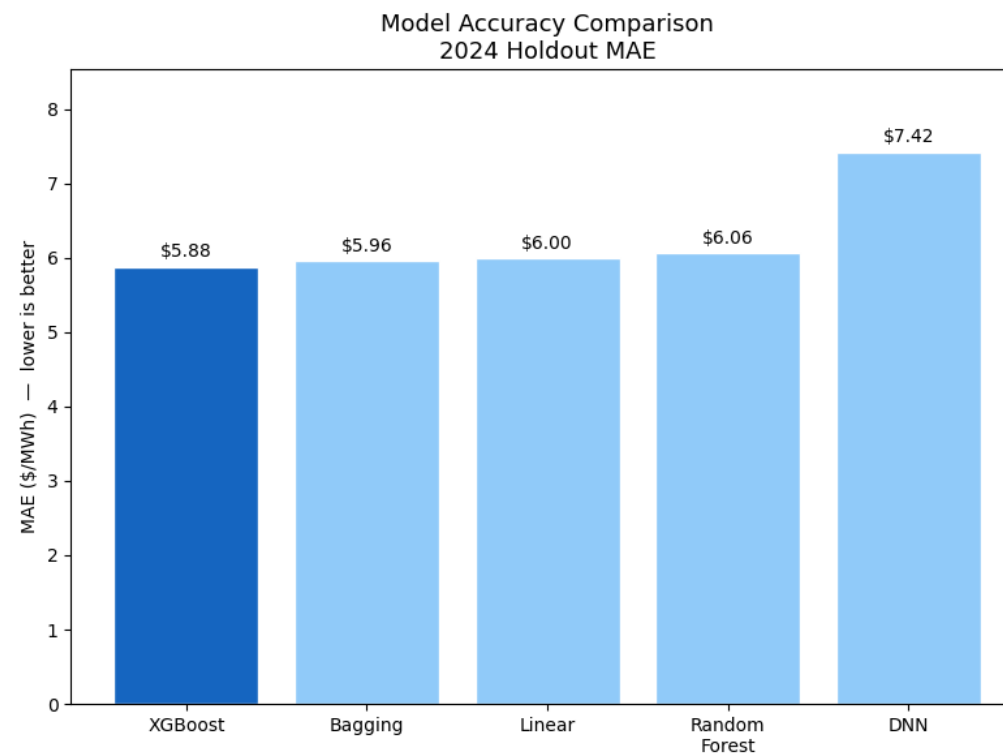
Linear, bagging, RF and XGBoost all land within \$0.20/MWh on holdout MAE.

- **Trees win on structure**

Native handling of mixed feature types, no scaling, robust to outliers with tuning.

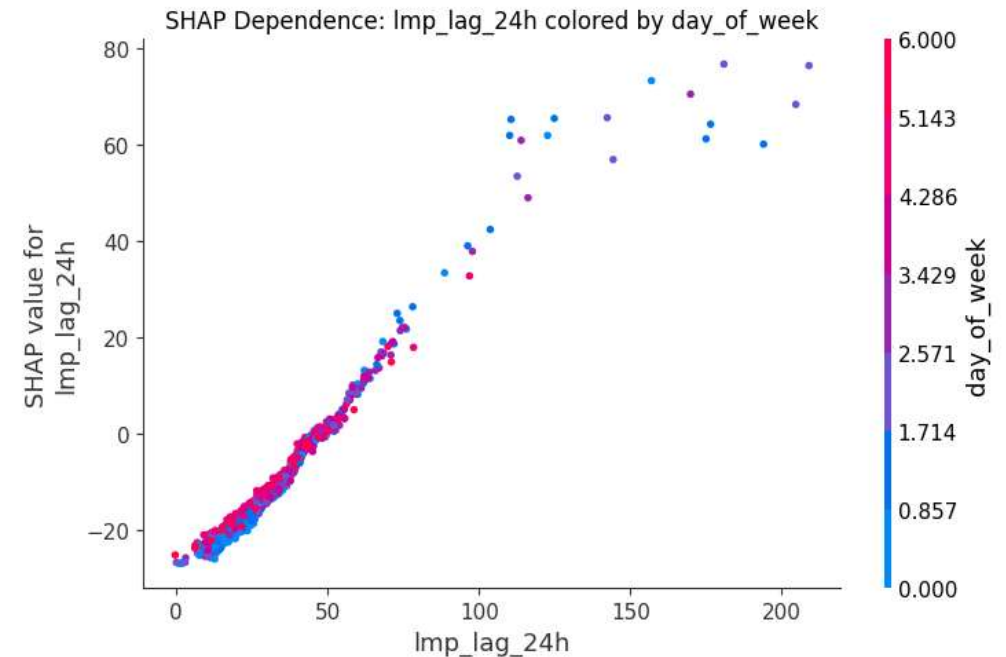
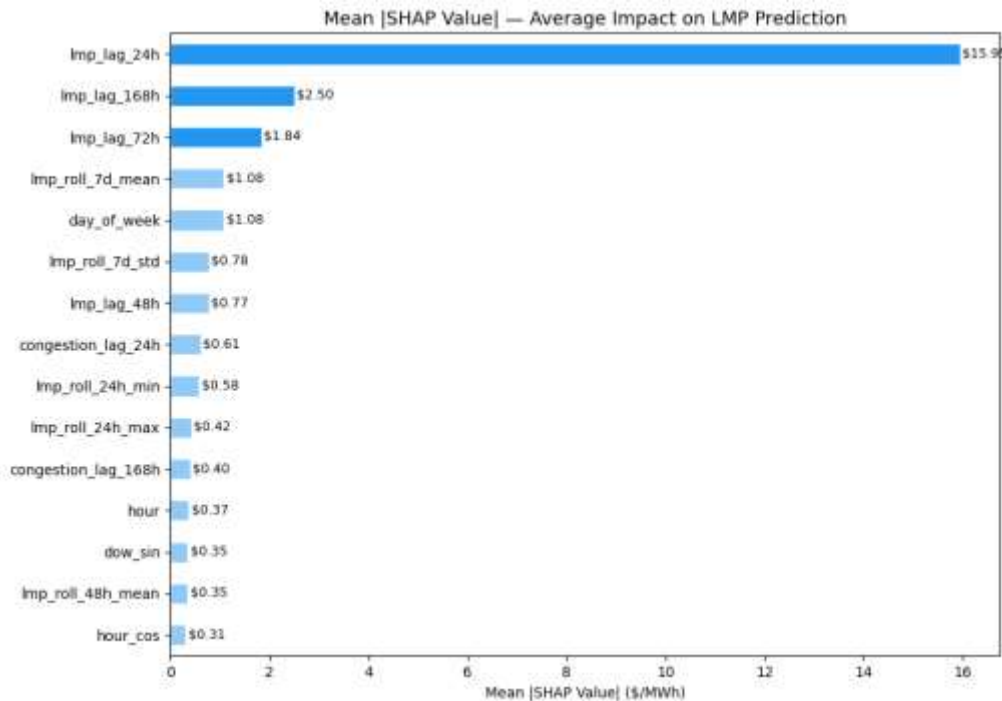
- **Interpretability matters**

Built-in feature importance + SHAP — essential for a client-facing, explainable model.



What drives the model?

One feature dominates: yesterday's price at the same hour. This is a persistence-driven market.

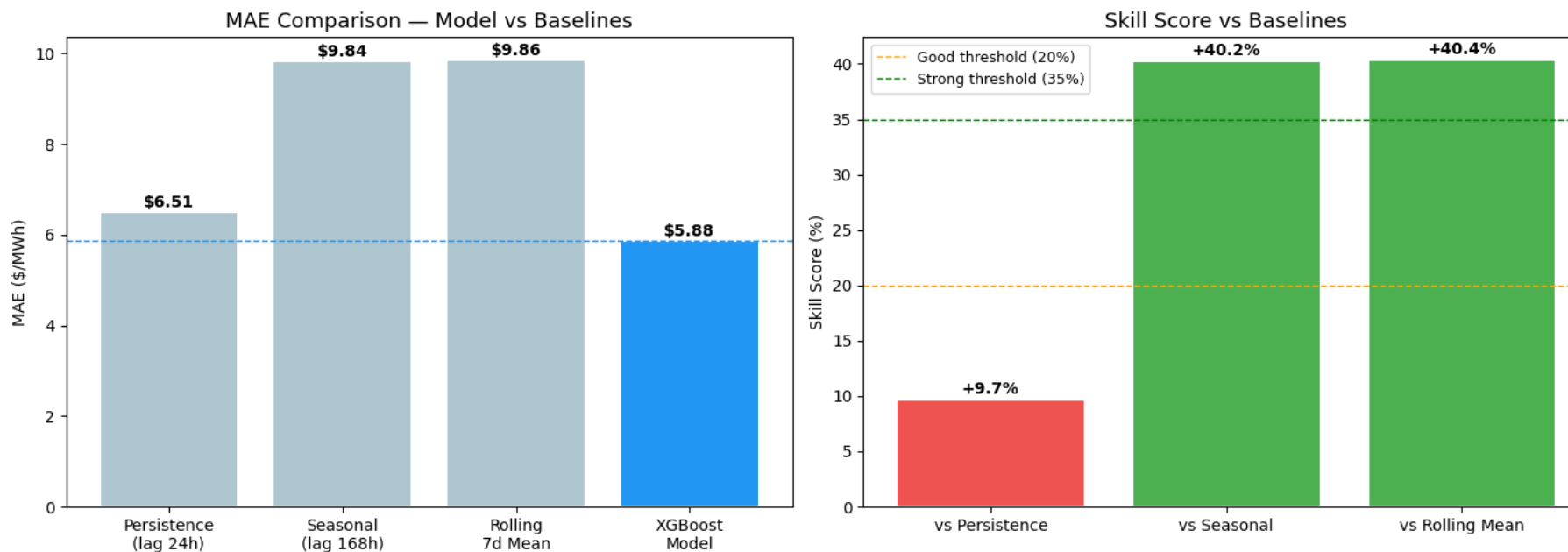


Mean |SHAP| impact (left): Imp_lag_24h $\approx$ \$15.95/MWh — every other feature is under \$2.50.

Dependence (right): the relationship is almost perfectly linear.

How good is it, really?

Beats weekly-seasonal and rolling-mean baselines by ~40%. But against naive 24-hour persistence: only +9.7%.

**+9.7%**

vs. persistence (lag-24h)

+40.2%

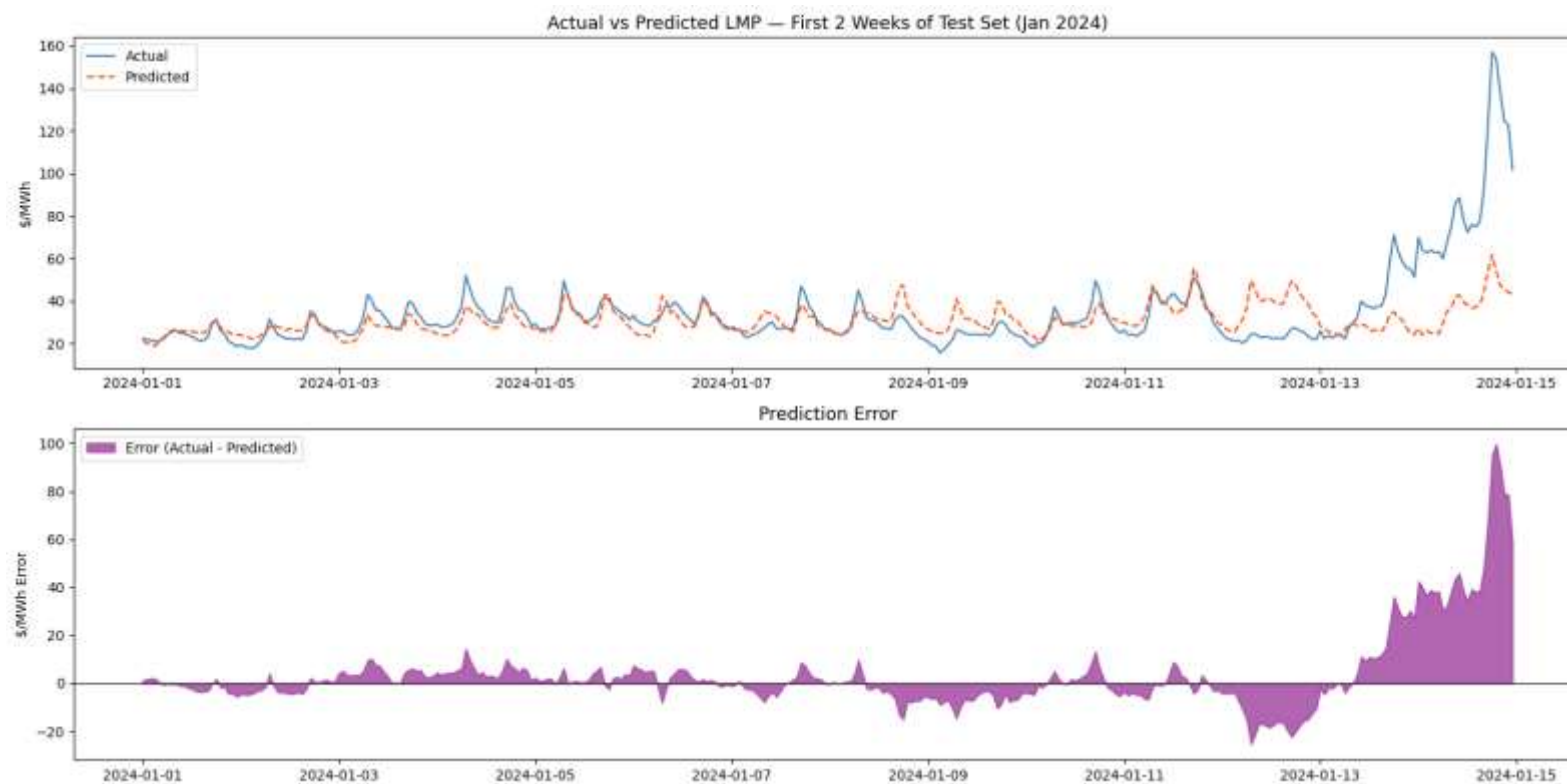
vs. weekly seasonal

+40.4%

vs. rolling mean

Where it breaks: January 2024

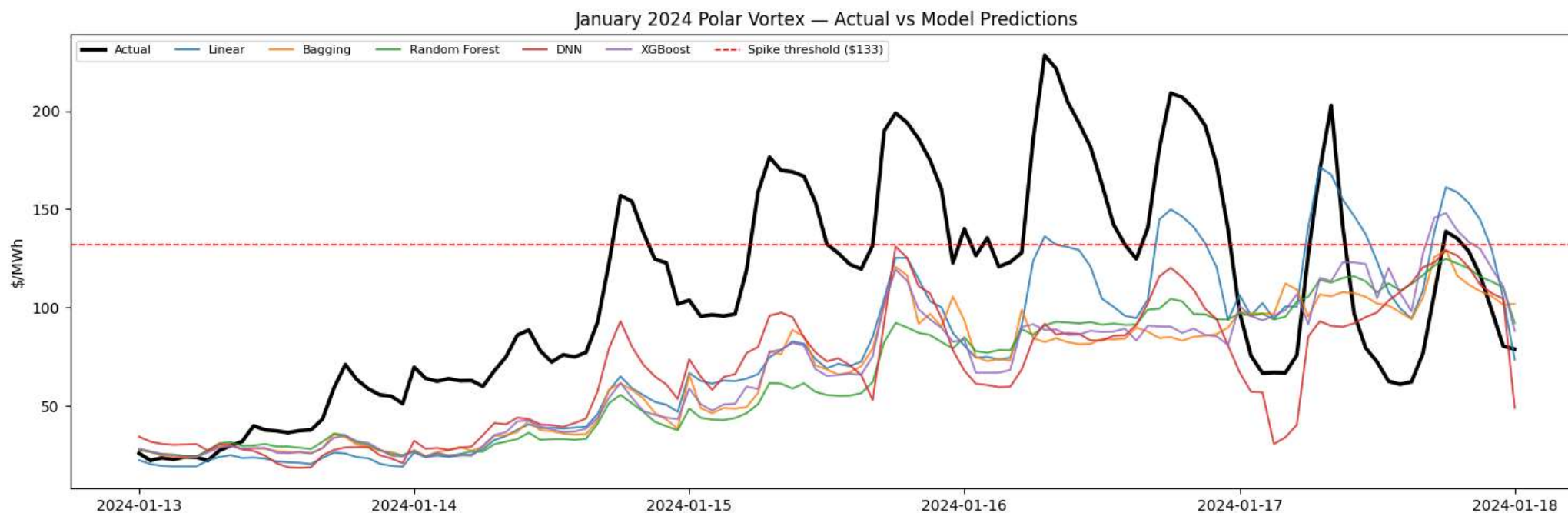
The model tracks the calm beautifully — then the polar vortex hits and it misses the crisis entirely.



First two weeks of the 2024 test set. Error stays near zero — until prices triple and the forecast flattens.

Every model fails the same way

Linear, bagging, random forest, neural net, XGBoost — all five miss the spikes together. The limit is structural, not algorithmic.



January 2024 polar vortex. Actual (black) blows past every model. No amount of normal-regime data fixes this — the price-formation mechanism itself changes during scarcity.

Strengths, limits, and the path forward

What can I do?

Learns the market from history. Beats ~10% over persistence and explains the normal market well.

Great for: dispatch, day-to-day positioning, expected revenue.



What can't I do?

Rare events structures differently, more factors than history price.

The model cannot predict something they've never seen in the past.



Where should I go?

- Add more features (weather from Open-Meteo, gas price from EIA)
- Separate models per LMP component (energy / congestion/loss)
- Structural simulation, model the system directly

We can sharpen the forecast or target the tail events.

Thank you

Questions & discussion welcome

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Walk-forward validation

Not a single split — 8 expanding folds. Early folds are inflated by thin training and the 2022 gas crisis; recent folds match the holdout.

